

JOHN BURKY

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PROFESSIONAL SUMMARY

Electrical Engineer in 4+1 master's program specializing in hardware design. Combines advanced research in precision electronics with hands-on experience in PCB layout, hardware verification, and RTL design. Available for full-time roles starting January 2027.

EDUCATION

Purdue University - West Lafayette (Online, part-time)

Master of Science, Electrical Engineering

January 2026 - December 2027

GPA: 4.0

Purdue University - West Lafayette

Bachelor of Science, Electrical Engineering

August 2023 - December 2026

GPA: 3.8

- Dean's List & Semester Honors: Fall 2023 - Spring 2026.

PROFESSIONAL EXPERIENCE

Birck Nanotechnology Center

Lab Assistant

November 2024 - Present

West Lafayette, IN, USA

- Engineered custom PCBs for high-precision resistor and capacitor characterization below 5 K, supporting semiconductor research in cryogenic environments and enabling repeatable laboratory measurements.
- Automated vacuum probe-station capacitance measurements in Python, reducing data-collection time by 50% and accelerating laboratory experiment cycles.
- Characterized lock-in amplifier response across filters and time constants and documented capacitor-measurement procedures, reducing experiment setup time and providing reference guidance for undergraduate researchers.

Purdue University - West Lafayette

Teaching Assistant (Introduction to C Programming)

January 2024 - Present

West Lafayette, IN, USA

- Taught weekly C-programming labs for 30 students and provided office-hour support on UNIX environments, memory management, and recursion; earned a 4.5/5.0 instructor rating across multiple semesters.
- Evaluated weekly labs and biweekly assignments, providing technical feedback while maintaining consistent grading standards.
- Mentored first-year teaching assistants on grading strategy and technical communication to support consistent student feedback.

Lake Shore Cryotronics

Electrical Engineering Intern

May 2024 - August 2026

Westerville, OH, USA

Full-time during summers; part-time during academic year

- Redesigned an ultra-low-noise JFET amplifier in Altium, reducing low-frequency noise by up to 65%; assembled the revised circuit and verified noise performance through hardware testing.
- Derived theoretical characteristic equations and developed parameter-fitting scripts to model proprietary cryogenic diode behavior, guiding the design of future device architectures.
- Interfaced a power management IC with an existing embedded system to enable a controlled shutdown sequence, preventing data corruption and improving system reliability.
- Authored comprehensive technical specifications and calibration procedures for 2 new product lines, supporting ISO 9001 compliance and streamlining the transition from prototype to production.
- Analyzed legacy instrument noise and improved shielding, reducing measurement noise by 45%.

PROJECTS & OUTSIDE EXPERIENCE

Multicore CPU Design

ECE 437: Computer Architecture

February 2026 - April 2026

West Lafayette, IN, USA

- Designed and verified a dual-core RISC-V processor in SystemVerilog with a multi-stage pipeline, integrated caches, and MSI cache coherence over a shared bus.
- Ranked 9th of 56 teams in mean program execution time; achieved 66 MHz operation on a Xilinx FPGA, exceeding the 50 MHz target by 32%.

SKILLS

Technical Skills: PCB Layout, Analog Design, Hardware Verification, Soldering, Low-Noise Design, RTL Design, I2C, SPI, UART

Languages: Java, C/C++, Python, RISC-V Assembly, SystemVerilog, MATLAB

Tools: Spectrum Analyzer, Oscilloscope, LTspice, Vivado, QuestaSim, Git, Linux, Altium